higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CONSTRUCTION PLANNING
NQF LEVEL 4

18 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1: CONSTRUCTION CONTRACTS

1.1
- The tender is advertised by the employer or employer's representative.
- Contractors visit the site to assess problems and difficulties that may affect the project.
- A contractor receives the tender document (bill of quantities) and pays a deposit.
- The tender document is completed with the relevant pricing.
- Contractors submit their completed tenders by the due date.
- Tenders are evaluated.
- The employer awards the contract to the winning tenderer. (Any 7 × 1) (7)

1.2
- Quantity surveyor
- Project manager
- Electrical engineer
- Health and safety consultant
- Architect
- Structural engineer
- Civil engineer
- Mechanical engineer
- Environmental consultant
- Fire consultant (Any 4 × 1) (4)

1.3
1.3.1 C
1.3.2 A
1.3.3 B
1.3.4 E

(4 × 1) (4)
[15]

QUESTION 2: BASIC COMMON LAWS APPLICABLE TO CONSTRUCTION

2.1 The neighbour's property may not be affected negatively by any work that is done on the building site. Excavation is required to build the foundations, and this may result in the loss of lateral support of the neighbouring property.✓
To avoid serious damage to the neighbouring property during the excavation process, it is required that artificial support is used.✓ The nature of the artificial support structure may cause cracking on the neighbour's property.✓
The owner of the land where the construction is taking place is obliged to ensure that there is uninterrupted lateral support to the neighbouring property.✓ (Any relevant 4 × 1) (4)

2.2
2.2.1 Common law
2.2.2 Contractor
2.2.3 Possession
2.2.4 Site
2.2.5 Employer
2.2.6 Pay

(6 × 1) (6)
[10]

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QUESTION 3: STANDARD CONSTRUCTION CONTRACTS

3.1 3.1.1 True
3.1.2 True
3.1.3 True
3.1.4 False
3.1.5 True

(5 × 1) (5)

3.2
- Daily site diary where incidents happening on the site are recorded
- Accurate minutes of all meetings on the site
- Proper records of instructions by the clerk of works/architect
- Accurate chart indicating progress of work (4 x 1) (4)

3.3 Joint Building Contracts Committee (1)

[10]

QUESTION 4: TENDERS AND QUOTES

4.1 4.1.1 Labour only contract
4.1.2 Open advertised tender
4.1.3 Restricted tender
4.1.4 Dayworks contracts
4.1.5 Lump sum contracts

(5 × 1) (5)

4.2
- Scope of work should give an accurate description of the building required.
- It should describe what is part and what is not part of the project.

The contractor should be able to tell from the scope of work:
- how big the project is;
- what skills and equipment are needed; and
- how long the project will take. (Any relevant 5 × 1) (5)

4.3
- The contractor's rates
- The current material cost
- The contractor's cost of labour
- The location of the site
- The workload of the contractor
- VAT inclusive or exclusive (Any 3 × 1) (3)

4.4
- Description column
- Unit column
- Quantity column
- Rate column
- Amount column
- Item reference (Any 2 × 1) (2)

[15]

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QUESTION 5: QUANTITIES AND COSTING

5.1 5.1.1 Centreline calculation:

2/8,000 = 16,000 ✓

2/4,320 = 8,640✓

24,640✓

less 4/0,220 - 0,880 ✓

23,760 x 2,600 = 61,776 m²✓✓

Gable wall: 4,320 x 1,800 = 7,776 m²✓

Total area: = 61,776 + 7,776 = 69,552 m²✓

Less Deductions: windows and door: (4 x 1,200 x 1,200) + (2 x 1,600 x 1,200) + (0,800 x 2,000)

= 11,200✓✓✓

69,552 – 11,200 = **58,352 m²**✓

No of bricks : 58,352 x 104 = **6068 no of bricks**✓✓ (14)

5.1.2 Cost of bricks = 6068 × R5,50
= R 33 374,00✓

Cost of mortar and labour = 58 × 165
= R9,570,00✓

Total cost = R33 374,00 + R9 570,00
= R42 944,00✓✓ (4)

5.2 Inside perimeter: 8,000 - 0,440 = 7,560
4,320 - 0,440 = 3,880

(7,560 + 3,880) x 2 = 22,880 m✓✓

(2)
[20]

QUESTION 6: PLANNING AND ORGANISING CONSTRUCTION ACTIVITIES

6.1 6.1.1 *Progressive sequencing* is doing construction activities one after another over time✓ in a logical order.✓ (2 × 1) (2)

6.1.2
- Fence off the site
- Provide containers and storage facilities
- Provide canteen and changing facilities
- Erect temporary toilets
- Install telephone lines (5 × 1) (5)

6.1.3
- Clear the site of all vegetation and boulders
- Set out the building
- Excavate foundation trenches
- Place concrete foundations
- Build foundation walls
- Backfill and compact soil within foundation walls
- Place concrete slabs
- Start with superstructure (Any 7 × 1 in the correct order) (7)

6.2 6.2.1 Critical activities
6.2.2 Non-critical activities
(2 × 1) (2)

6.3 6.3.1 Critical activity
6.3.2 Non-critical activity
6.3.3 Critical activity
6.3.4 Non-critical activity
(4 × 1) (4)
[20]

QUESTION 7: PROJECTIONS AND CASH FLOW

7.1
- Project starting date
- Activities in their correct sequence
- Duration in calendar days, including weekends
- Public holidays and end of year builders' holidays
- Handover date
- Final completion date
- Delays (Any 4 × 1) (4)

7.2 7.2.1 Payment to be received at the end of second month
= R45 000✓ + (15%) R6 750,00✓ = R51 750,00✓

7.2.2 Payment to be received at the end of third month
= R51 750✓+ (15%) R7 762,50✓ = R59 512,50✓
(Alternative methods are accepted) (2 × 3) (6)
[10]

TOTAL: 100

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